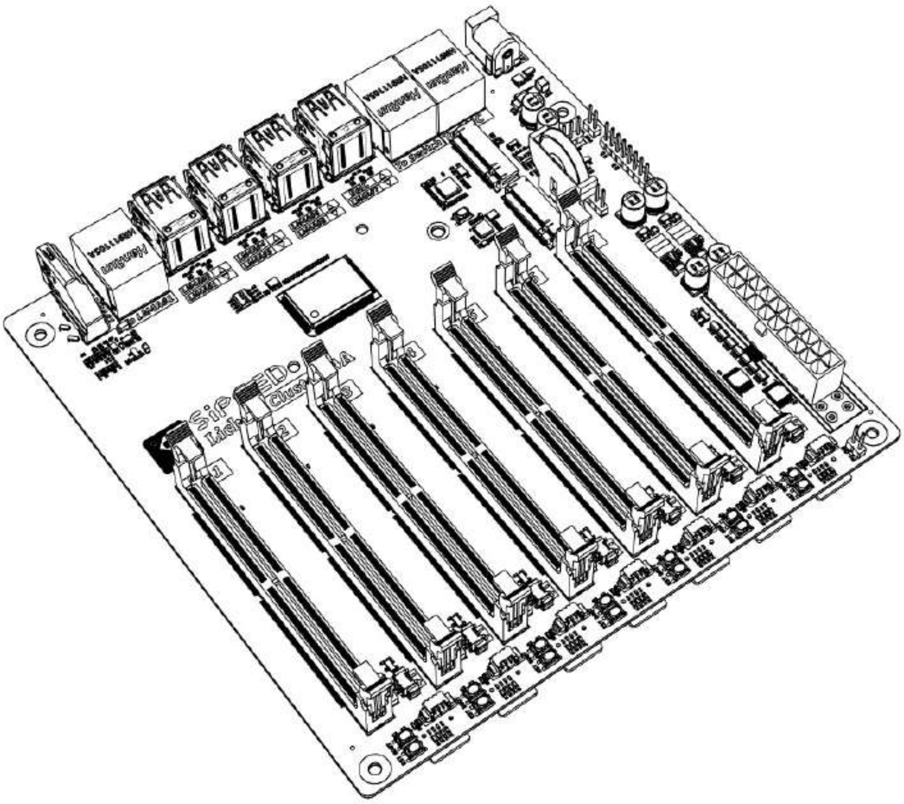
LC4A DataSheet

V1.0



Revision History

Date	Revision	Description
2023-05-04	1.0	Initial Release

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Contents

Overview

The Lichee Cluster 4 A (LC4A) is a 7 nodes cluster on a mini ITX board. It’ s a scale model of bare metal clusters like you see in data centers. The LC4A board can be used as a local mini server to host apps, containers, Kubernetes. The LC4A also is a great tool for learning cloud-native technologies, Serverless, Microservices on a bare-metal cluster. The LC4A top/main node can act as a NAT/Router for the rest of the nodes. If you move the cluster from one location to another, all the node's IPs stay the same.

LC4A uses a standard ITX form factor, you can use any common chassis that supports ITX form factor, and supports ATX power supply

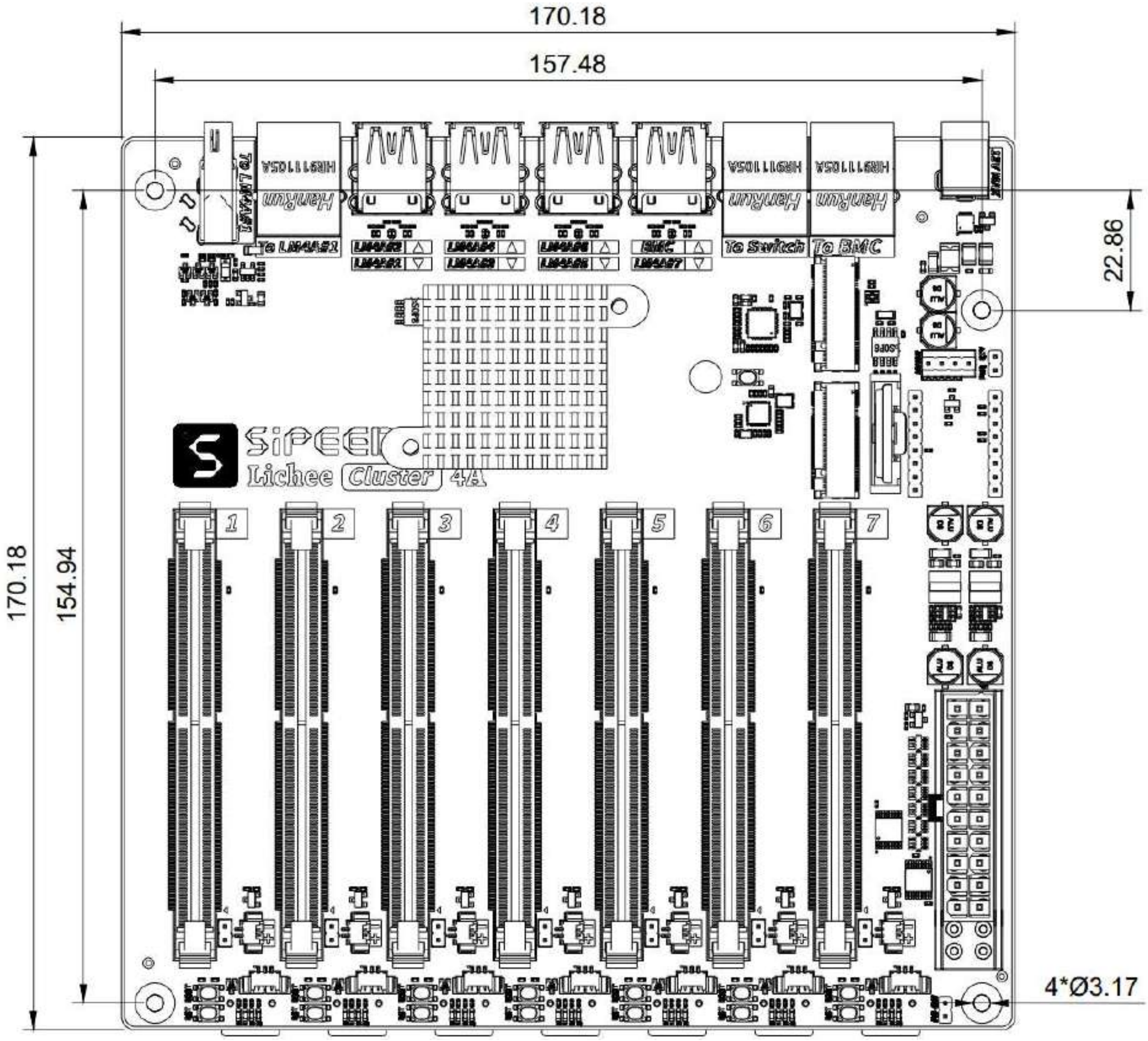
1 Key Specifications

Parameter	Description
SOM	7 x LM4A SOM Board
LM4A CPU	TH1520 RISC-V 4 x C910@1.85GHz
LM4A GPU	OpenGL ES 3.2 OpenCL 1.2 Vulkan 1.2
LM4A NPU	4TOPS@INT8
LM4A Memory	8GB/16GB LPDDR4X
LM4A Storage	TF card/eMMC
BMC	Lichee RV SOM
Video	1 x HDMI OUT (Connect to LM4A#1)
Ethernet	1 x 1000M ETH (Connect to LM4A#1) 1 x 1000M ETH (Connect to Switch) 1 x 100M ETH (Connect to BMC)
USB	7 x USB Type-A 3.0 HOST (Connect to LM4) 1 x USB Type-A 2.0 HOST (Connect to BMC) 1 x USB Type-C (Only For PD POWER)
Power Input	DC : 12V Max ATX : Support PC ATX 24Pin Power Don't use them at the same time
POWER Management	Support SIPEED M0 Sense (BouffaloLab BL702)
OS	LM4A : Debian, deepin, ubuntu, openwrt BMC : OpenBMC
Size	17*17CM ITX

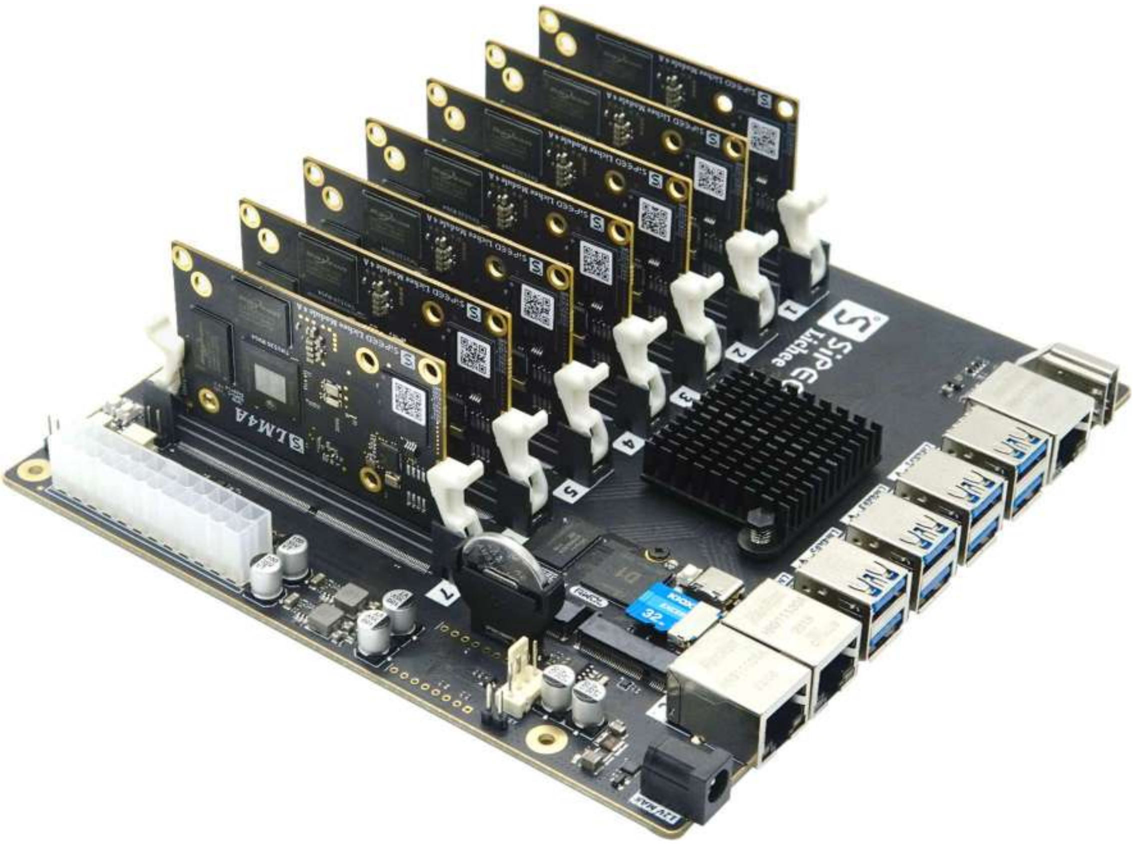
2 Dimensions and Interfaces

2.1 Size and Structure

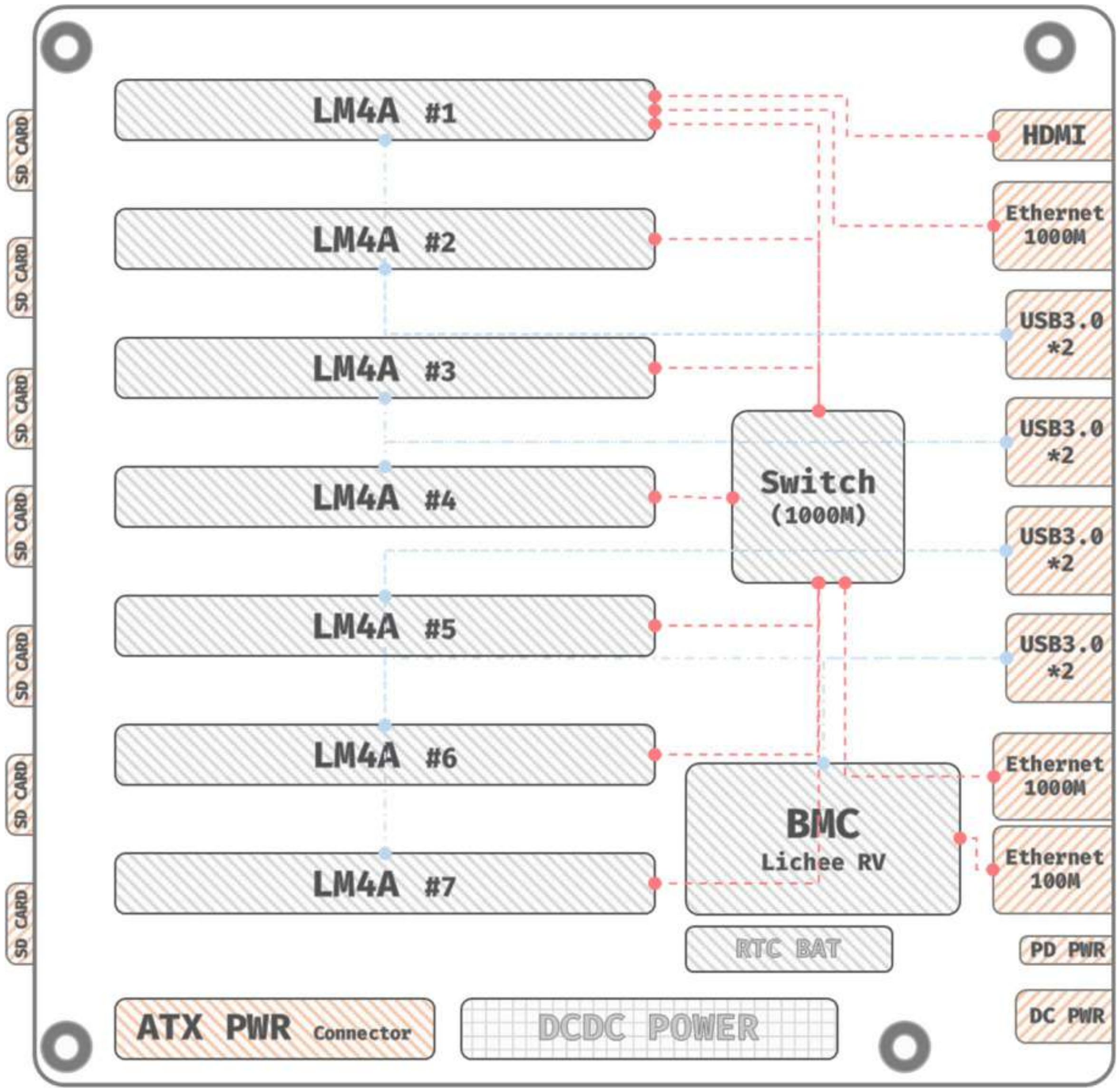
The lc4A is compatible with standard ITX form factors.

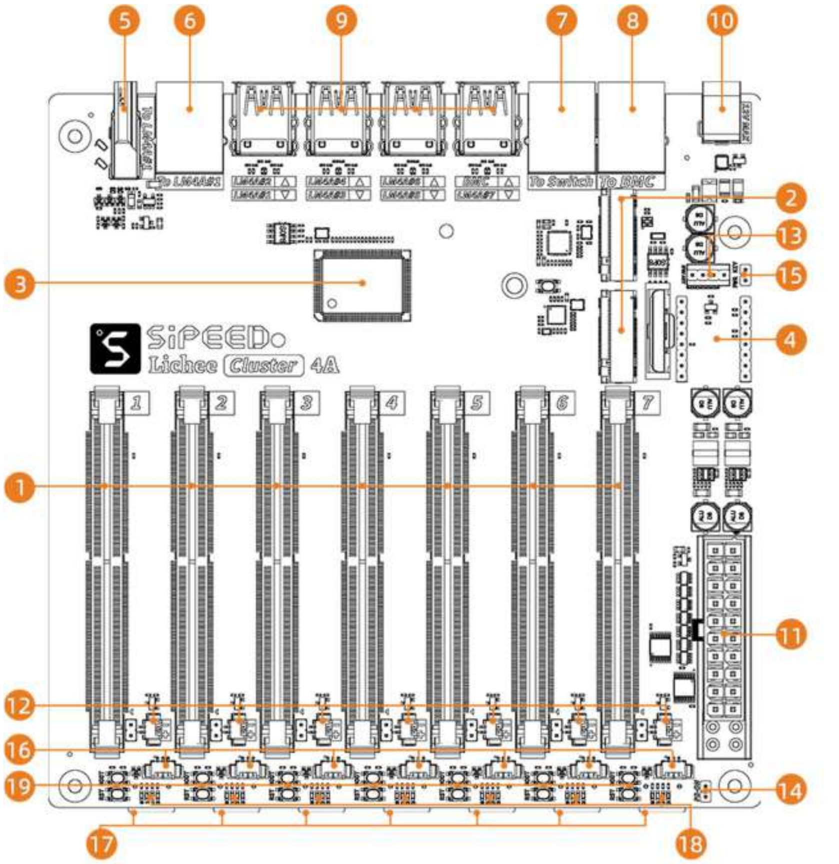


2.2 Physical Picture



2.3 Function Description





1. SO-DIMM 260P slots * 7
2. BMC slot (Lichee RV SOM)
3. Switch Chip RTL8370N
4. Power management module reserved bit, support Sipeed M0 Sense
5. HDMI Connector
6. Gigabit Ethernet connector (slot #1)
7. Gigabit Ethernet connectors (switches)
8. 100Gb Ethernet Connector (BMC)
9. USB3.0 * 7 + USB2.0 * 1
10. DC power input socket (12V max)
11. ATX 24Pin Power Socket
12. 5V fan connector * 7
13. 12V PWM fan connector * 1
14. ATX Power Switch Shorting Cap
15. Chassis switch button interface
16. Serial debug port * 7
17. TF card holders * 7
18. Indicator lights * 7 groups
19. BOOT and RST buttons * 7

3 Contact and information

Item	Description
Documentation	wiki.sipeed.com/licheecluster4a
QQ	559614960
Telergram	https://t.me/linux4rv
BBS	maixhub.com/discussion
Business Email	support@sipeed.com

4 Operating Precautions

During the use of this product, attention must be paid to operation safety and maintenance, otherwise it may cause damage to the product, shorten its service life, and even endanger personal safety. For safe use and maintenance, attention should be paid to the following aspects:

- This product is a high-precision electronic product. Please do not collide or fall.
- Although the Class1 laser used in this product meets the safety standards for human eyes, it is not recommended to look directly at the laser for a long time to avoid discomfort.
- Do not place this product in a place with high temperature or direct sunlight.
- Do not disassemble or modify this product without permission to prevent damage to the components of the product.
- Do not touch the product to avoid leaving fingerprints and other pollutants affecting the image effect.
- Please keep this product out of the reach of children to prevent accidents.
- Please follow the manual for correct and safe operation.